2nd Order DEs: Homogeneous

Questions 1–34

Question 1. For each of the following differential equations:

- state its order,
- state whether it is linear,
- state whether it is homogeneous,
- determine whether an auxiliary equation can be formed.

a) $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 4y = 0$
b) $y'' + y^2 = 0$
c) $3y'' - 5y' = 7$
d) $y''' + 2y' - y = 0$
e) $\frac{d^2y}{dx^2} + x\frac{dy}{dx} - 2y = 0$
f) $\ddot{x} + 4\dot{x} + 4x = e^x$
g) $(y')^2 + y = 0$
h) $2\ddot{x} - 3\dot{x} + x = 0$

Question 2. It is given that $\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 10y = 0$. Verify that these are all particular solutions of the differential equation:

a) $y = e^{2x}$
b) $y = 4e^{2x}$
c) $y = e^{5x}$
d) $y = 7e^{5x}$
e) $y = e^{2x} + e^{5x}$
f) $y = 4e^{2x} + 7e^{5x}$

Question 3. It is given that $\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 10y = 0$. Let $y = e^{mx}$, where m is a constant.

a) Show that $m^2 - 7m + 10 = 0$.

The quadratic equation has roots m_1 and m_2.

a) Show that $y = Ae^{m_1 x} + Be^{m_2 x}$ is a solution to the differential equation, where A and B are constants.

Question 4. For each of the following differential equations, by forming and solving the auxiliary equation, find the general solution.

a) $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$

b) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 35y = 0$

c) $\frac{d^2y}{dx^2} + 18\frac{dy}{dx} + 80y = 0$

d) $2\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0$

e) $\frac{d^2y}{dx^2} - 4y = 0$

f) $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} = 0$

Question 5. Find the general solution to the following differential equations.

a) $\ddot{x} + 3\dot{x} - 28x = 0$

b) $4\ddot{x} + 4\dot{x} - 15x = 0$

Question 6. Solve the differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = 0$ given that $y = 2$ when $x = 0$, and $y = 3$ when $x = \ln 2$.

Question 7. Solve the differential equation $2y'' - 7y' + 3y = 0$ given that $y = 5$ when $x = 0$, and $y = -238$ when $x = \ln 4$.

Question 8. It is given that $\dfrac{d^2y}{dx^2} - 6\dfrac{dy}{dx} + 9y = 0$. Verify that these are all particular solutions of the differential equation:

a) $y = e^{3x}$
b) $y = 2e^{3x}$
c) $y = xe^{3x}$
d) $y = 5xe^{3x}$
e) $y = e^{3x} + xe^{3x}$
f) $y = 2e^{3x} + 5xe^{3x}$

Question 9. It is given that $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$. Let $y = xe^{mx}$, where m is a constant.

a) Show that $m^2 - 6m + 9 = 0$.

The quadratic equation has a repeated root m.

a) Show that $y = Axe^{mx} + Be^{mx}$ is a solution to the differential equation, where A and B are constants.

Question 10. For each of the following differential equations, by forming and solving the auxiliary equation, find the general solution.

a) $\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = 0$

b) $\frac{d^2y}{dx^2} + 12\frac{dy}{dx} + 36y = 0$

c) $4\frac{d^2y}{dx^2} + 12\frac{dy}{dx} + 9y = 0$

d) $9\frac{d^2y}{dx^2} - 42\frac{dy}{dx} + 49y = 0$

Question 11. Find the general solution to the following differential equations.

a) $y'' + 26y' + 169y = 0$

b) $9y'' - 48y' + 64y = 0$

Question 12. Solve the differential equation $\ddot{x} - 10\dot{x} + 25x = 0$ given that $x = 0$ when $t = 0$, and $x = 4$ when $t = 1$.

Question 13. Solve the differential equation $9\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + y = 0$ given that $y = 3$ when $x = 3$, and $y = 4$ when $x = 6$.

Question 14. It is given that $\frac{d^2y}{dx^2} + 4y = 0$. You may assume that $\frac{d}{dx}(e^{ikx}) = ke^{ikx}$, where k is a constant. Verify that these are all particular solutions of the differential equation:

a) $y = e^{2ix}$
b) $y = e^{-2ix}$
c) $y = 2e^{2ix} - 3e^{-2ix}$

Question 15. Use $e^{i\theta} = \cos\theta + i\sin\theta$ to show that $Ae^{2ix} + Be^{-2ix}$ can be written in the form $C\cos 2x + D\sin 2x$, where C and D are real constants.

Question 16. It is given that $\dfrac{d^2y}{dx^2} + 4y = 0$. Verify that these are all particular solutions of the differential equation:

a) $y = \cos 2x$
b) $y = \sin 2x$
c) $y = 3\cos 2x$
d) $y = 5\sin 2x$
e) $y = \cos 2x + \sin 2x$
f) $y = 3\cos 2x + 5\sin 2x$

Question 17. It is given that $\frac{d^2y}{dx^2} + 9y = 0$. Let $y = e^{mx}$, where m is a constant.

a) Show that $m^2 + 9 = 0$.

The quadratic equation has complex conjugate roots.

a) Show that $y = A\cos 3x + B\sin 3x$ is a solution to the differential equation, where A and B are constants.

Question 18. For each of the following differential equations, by forming and solving the auxiliary equation, find the general solution.

a) $\frac{d^2y}{dx^2} + 25y = 0$

b) $\frac{d^2y}{dx^2} + 49y = 0$

c) $4\frac{d^2y}{dx^2} + 9y = 0$

Question 19. Find the general solution to the following differential equations.

a) $\ddot{x} + 8x = 0$

b) $9\ddot{x} + 25x = 0$

Question 20. Solve the differential equation $y'' + 100y = 0$ given that $y = 0$ when $x = 0$, and $y = 3$ when $x = \dfrac{\pi}{20}$.

Question 21. Solve the differential equation $16\dfrac{d^2y}{dx^2} + y = 0$ given that $y = 2$ when $x = \pi$, and $y = -3$ when $x = 2\pi$.

Question 22. It is given that $\dfrac{d^2y}{dx^2} - 4\dfrac{dy}{dx} + 13y = 0$. Let $y = e^{mx}$, where m is a constant.

a) Show that $m^2 - 4m + 13 = 0$.

b) Hence show that the roots of this equation can be written in the form $a \pm bi$.

Question 23. It is given that $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$. Verify that these are all particular solutions of the differential equation:

a) $y = e^{(2+3i)x}$

b) $y = e^{(2-3i)x}$

c) $y = 5e^{(2+3i)x} + 2e^{(2-3i)x}$

Question 24. Use $e^{i\theta} = \cos\theta + i\sin\theta$ to show that $Ae^{(2+3i)x} + Be^{(2-3i)x}$ can be written in the form $e^{2x}(C\cos 3x + D\sin 3x)$, where C and D are real constants.

Question 25. It is given that $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$. Verify that these are all particular solutions of the differential equation:

a) $y = e^{2x}\cos 3x$
b) $y = e^{2x}\sin 3x$
c) $y = 4e^{2x}\cos 3x$
d) $y = 5e^{2x}\sin 3x$
e) $y = e^{2x}\cos 3x + e^{2x}\sin 3x$
f) $y = 4e^{2x}\cos 3x + 5e^{2x}\sin 3x$

Question 26. It is given that $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 29y = 0$. Let $y = e^{mx}$, where m is a constant.

a) Show that $m^2 - 4m + 29 = 0$.
b) Hence show that the roots of this equation can be written in the form $a \pm bi$.
c) Show that $y = e^{2x}(A\cos 5x + B\sin 5x)$ is a solution to the differential equation, where A and B are constants.

Question 27. For each of the following differential equations:

- Form the auxiliary equation,
- Solve the auxiliary equation,
- Write down the general solution.

a) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 5y = 0$

b) $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 13y = 0$

c) $\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 25y = 0$

Question 28. Find the general solution to the following differential equations.

a) $y'' - 6y' + 73y = 0$

b) $16y'' - 80y' + 101y = 0$

Question 29. Solve the differential equation $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 10y = 0$ given that $y = 4$ when $x = 0$, and $y = 2$ when $x = \frac{\pi}{2}$.

Question 30. Solve the differential equation $\ddot{x} - 4\dot{x} + 13x = 0$ given that $x = 2$ when $t = 0$, and $\dot{x} = 1$ when $t = 0$.

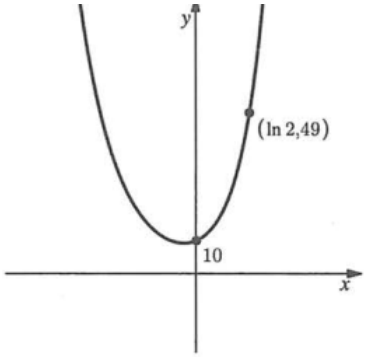
Question 31. Find the general solution to the following differential equations.

a) $\dfrac{d^2y}{dx^2} - 10\dfrac{dy}{dx} + 16y = 0$
b) $\dfrac{d^2y}{dx^2} - 18\dfrac{dy}{dx} + 97y = 0$
c) $\ddot{x} - 11\dot{x} - 60x = 0$
d) $121y'' - 110y' + 25y = 0$
e) $9\ddot{x} + 42\dot{x} + 49x = 0$
f) $\dfrac{d^2y}{dx^2} - 30\dfrac{dy}{dx} + 225y = 0$
g) $\dfrac{d^2y}{dx^2} + y = 0$
h) $2\ddot{x} + 2\dot{x} + 5x = 0$
i) $\dfrac{d^2y}{dx^2} + 6\dfrac{dy}{dx} + 109y = 0$
j) $81y'' + y = 0$
k) $\dfrac{d^2y}{dx^2} + 5\dfrac{dy}{dx} - 36y = 0$
l) $\dfrac{d^2y}{dx^2} + 16\dfrac{dy}{dx} + 64y = 0$
m) $49\ddot{x} + 25x = 0$
n) $30y'' + 49y' + 20y = 0$
o) $\dfrac{d^2y}{dx^2} + 400y = 0$
p) $\ddot{x} + 32x = 0$
q) $32y'' - 12y' - 35y = 0$

Question 32. For each of the following functions y, find the simplest 2nd order linear homogeneous differential equation with constant coefficients, for which the given function is a solution.

a) $y = 5e^{-3x} + 6e^{4x}$
b) $y = e^{6x}(2\cos 5x + 9\sin 5x)$
c) $y = 6xe^{14x} - 5e^{14x}$
d) $y = 15\cos^2 \frac{2}{3}x + 6\sin^2 \frac{2}{3}x$

Question 33. It is given that $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 0$. A solution to the differential equation is shown in the diagram below. It is given that the curve passes through $(0, 10)$ and $(\ln 2, 49)$. Find the equation of the curve.



Question 34. It is given that $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$. A solution to the differential equation is shown in the diagram below. It is given that the curve passes through $(0, 2)$ and that the curve has a stationary point when $x = \frac{\pi}{6}$. Find the equation of the curve.

